Unit 0.01: EST I Literacy & Quantitative Diagnostic Drill

Q1. Consider the functions fx = 2x + 3 and gx = x + 7 . At what x -value do the graphs of fx and gx intersect?

[x] A: 4
[ ] B: 7
[ ] C: 3
[ ] D: 10

Solution: To find the x-value where the graphs of fx and gx intersect, we set the two functions equal to each other: fx = gx . nnSubstituting the given expressions for fx and gx : n 2x + 3 = x + 7 nnTo solve for x , subtract x from both sides of the equation: n 2x - x + 3 = x - x + 7 nnThis simplifies to: n x + 3 = 7 nnNext, subtract 3 from both sides of the equation: n x + 3 - 3 = 7 - 3 nnWhich gives the solution: n x = 4 nnTherefore, the graphs of fx and gx intersect at x = 4 .

Q2. The graph of the quadratic function fx = x^2 - 4x + 3 is shown below. What is the value of f0 ?

[ ] A: -3
[ ] B: -1
[ ] C: 1
[x] D: 3

Solution: To find the value of f0 , we substitute x = 0 into the function's equation: fx = x^2 - 4x + 3 . Therefore, f0 = 0^2 - 40 + 3 = 0 - 0 + 3 = 3 . This corresponds to the y-intercept of the graph.

Q3. In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle angle AGH is 125^circ, what is the measure of angle angle GHD? text{Diagram description:} text{ Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G is on AB, H is on CD. Angle AGH is in the upper left quadrant formed by the intersection.}

[ ] A: 55^circ
[x] B: 125^circ
[ ] C: 90^circ
[ ] D: 180^circ

Solution: When two parallel lines are intersected by a transversal, consecutive interior angles are supplementary add up to 180^circ. Angle angle AGH and angle angle GHC are consecutive interior angles. Therefore, angle AGH + angle GHC = 180^circ. We are given that angle AGH = 125^circ. So, 125^circ + angle GHC = 180^circ. Solving for angle GHC, we get angle GHC = 180^circ - 125^circ = 55^circ. Angle angle GHC and angle angle GHD are angles on a straight line linear pair, so they are supplementary. Therefore, angle GHC + angle GHD = 180^circ. Substituting the value of angle GHC, we get 55^circ + angle GHD = 180^circ. Solving for angle GHD, we get angle GHD = 180^circ - 55^circ = 125^circ. Alternatively, angle angle AGH and angle angle GHD are alternate exterior angles. When two parallel lines are intersected by a transversal, alternate exterior angles are equal. Therefore, angle AGH = angle GHD. Since angle AGH = 125^circ, then angle GHD = 125^circ.

Unit 0.01: EST I Literacy & Quantitative Diagnostic Drill

Q4. The bar chart below shows the number of students who prefer each of four different colors. What is the total number of students who prefer either blue or green?

[ ] A: 30
[x] B: 40
[ ] C: 50
[ ] D: 60

Solution: The bar chart shows the following counts: Red: 20 students, Blue: 25 students, Green: 15 students, Yellow: 10 students. To find the total number of students who prefer either blue or green, we add the number of students who prefer blue to the number of students who prefer green. Total = Number of students who prefer blue + Number of students who prefer green Total = 25 + 15 Total = 40 Therefore, 40 students prefer either blue or green.

Q5. Consider the functions fx = 2x + 3 and gx = x + 7 . At what x -value do the graphs of fx and gx intersect?

[x] A: 4
[ ] B: 7
[ ] C: 3
[ ] D: 10

Solution: To find the x-value where the graphs of fx and gx intersect, we set the two functions equal to each other: fx = gx . nnSubstituting the given expressions for fx and gx : n 2x + 3 = x + 7 nnTo solve for x , subtract x from both sides of the equation: n 2x - x + 3 = x - x + 7 nnThis simplifies to: n x + 3 = 7 nnNext, subtract 3 from both sides of the equation: n x + 3 - 3 = 7 - 3 nnWhich gives the solution: n x = 4 nnTherefore, the graphs of fx and gx intersect at x = 4 .

Q6. A recipe calls for 2 cups of flour for every 3 cups of sugar. If 8 cups of flour are used, how many cups of sugar are needed?

[ ] A: 10 cups
[ ] B: 11 cups
[x] C: 12 cups
[ ] D: 13 cups

Solution: Let F be the amount of flour and S be the amount of sugar. The ratio of flour to sugar is given as frac{F}{S} = frac{2}{3} . We are given that F = 8 cups. We need to find S . We can set up the proportion: frac{8}{S} = frac{2}{3} . To solve for S , we can cross-multiply: 8 times 3 = 2 times S . This simplifies to 24 = 2S . Dividing both sides by 2, we get S = frac{24}{2} = 12 . Therefore, 12 cups of sugar are needed.

Q7. If 9^{x+1} = 27^{x-1} , what is the value of x ?

[ ] A: 2
[ ] B: 3
[x] C: 5
[ ] D: 7

Solution: To solve the equation 9^{x+1} = 27^{x-1} , express both sides in terms of a common base: 1. Recognize that both 9 and 27 are powers of 3 : 9 = 3^2 quad text{and} quad 27 = 3^3 2. Substitute these into the original equation: 3^2^{x+1} = 3^3^{x-1} 3. Use the power of a power rule, a^m^n = a^{mn} , distributing across both terms in each exponent: 3^{2x+1} = 3^{3x-1} 3^{2x+2} = 3^{3x-3} 4. Since the bases are equal and positive and not equal to 1, equate the exponents: 2x + 2 = 3x - 3 2 + 3 = 3x - 2x x = 5 Common misconceptions: - Forgetting to distribute the factor to the constant term e.g., writing 2x + 1 = 3x - 1 yields x = 2 . - Treating the bases as linear multipliers e.g., 9x+1 = 27x-1 also incorrectly leads to x = 2 .

Unit 0.01: EST I Literacy & Quantitative Diagnostic Drill

Q8. The graph of the quadratic function fx = x^2 - 4x + 3 is shown below. What is the value of f0 ?

[ ] A: -3
[ ] B: -1
[ ] C: 1
[x] D: 3

Solution: To find the value of f0 , we substitute x = 0 into the function's equation: fx = x^2 - 4x + 3 . Therefore, f0 = 0^2 - 40 + 3 = 0 - 0 + 3 = 3 . This corresponds to the y-intercept of the graph.